

FITSENSE AI

Personalized Fitness Recommendations Powered by Artificial Intelligence

NAAN MUDHALVAN PROJECT DOCUMENTATION

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B.Sc. Computer Science with Artificial Intelligence

1. Abstract

FitSense AI is an AI-powered fitness management system developed as part of the Naan Mudhalvan project. It allows authenticated users to record and manage workouts and receive personalized fitness recommendations and AI-generated fitness insights. The backend uses Node.js, Express.js, MongoDB, JWT authentication, bcrypt.js, and Google Gemini AI.

2. Introduction

FitSense AI combines workout tracking with Artificial Intelligence. Users can securely store workout history and obtain recommendations and insights based on fitness goals, experience, and activity statistics.

3. Problem Statement

Basic fitness trackers mainly store workout information and provide limited personalization. Users need a secure system that can manage workout data and provide intelligent, personalized guidance.

4. Objectives

- Secure user registration and login using JWT.
- Create, view, search, update, and delete workout records.
- Store user and workout data in MongoDB.
- Generate personalized workout recommendations using Gemini AI.
- Generate AI-based fitness insights from workout statistics.
- Provide a structured REST API for future frontend integration.

5. Existing System

Traditional fitness tracking systems focus mainly on recording workout activity. Users often need to interpret their own statistics or use separate sources for personalized guidance.

6. Proposed System

FitSense AI integrates workout management with AI assistance. Authenticated users can manage workouts and send fitness information to Gemini AI, which returns personalized recommendations and insights through REST APIs.

7. Technology Stack

- Backend: Node.js, Express.js
- Database: MongoDB, Mongoose
- Security: JWT, bcrypt.js
- Artificial Intelligence: Google Gemini AI
- API Testing: Postman
- IDE: Visual Studio Code

8. System Architecture

Client/Postman → Express Server → Routes → Authentication Middleware → Controllers → Services → MongoDB/Gemini AI. Routes receive requests, middleware validates authentication, controllers process application logic, services communicate with AI, and MongoDB stores application data.

9. Modules

9.1 Authentication Module

Provides registration and login. Passwords are hashed before storage and successful login returns a JWT token for protected APIs.

9.2 Workout Management Module

Supports creating, viewing, searching, updating, and deleting workout records. Each workout contains workout name, category, duration, calories burned, workout date, and the associated user.

9.3 AI Workout Recommendation Module

Accepts age, fitness goal, and experience level. Gemini AI generates personalized workout planning, weekly suggestions, level-based recommendations, motivation, and safety considerations.

9.4 AI Fitness Insights Module

Accepts total workouts, average duration, and total calories burned. Gemini AI generates performance observations, improvement suggestions, motivation, and a progress summary.

10. Database Design

The User model contains name, email, hashed password, and timestamps. The Workout model contains user reference, workout name, category, duration, calories burned, workout date, and timestamps.

11. API Documentation

Module	Method	Endpoint	Purpose
Authentication	POST	/api/auth/register	Register user
Authentication	POST	/api/auth/login	Login and receive JWT
Authentication	GET	/api/auth/profile	Get profile
Workouts	POST	/api/workouts	Create workout
Workouts	GET	/api/workouts	Get workouts
Workouts	GET	/api/workouts/:id	Get workout by ID
Workouts	PUT	/api/workouts/:id	Update workout
Workouts	DELETE	/api/workouts/:id	Delete workout
Workouts	GET	/api/workouts/search?q=	Search workouts
AI	POST	/api/ai/workout-recommendation	Generate recommendation
AI	POST	/api/ai/fitness-insights	Generate fitness insights

12. Security

- JWT authentication protects user-specific resources.
- bcrypt.js hashes passwords before storage.

- Sensitive configuration is stored in environment variables.
- The .env file is excluded from Git using .gitignore.

13. Testing and Results

The application was tested using Postman. Authentication, workout creation and retrieval, search, update/delete operations, AI workout recommendations, and AI fitness insights were tested using JSON requests and responses. The backend health endpoint and MongoDB connection were also verified.

14. Demo Video

Project demonstration video:

<https://drive.google.com/file/d/1BF0S1bmLlihAAHe3QE5Qda9EAtFPm06K/view?usp=drivesdk>

15. Advantages

- Combines workout tracking and AI assistance.
- Provides personalized recommendations.
- Supports secure authentication and user-specific data.
- Uses modular MVC architecture.
- Can be extended with a web or mobile frontend.

16. Future Enhancements

- Responsive web/mobile frontend.
- Progress dashboards and charts.
- Goal tracking and notifications.
- Wearable/activity tracker integration.
- Expanded AI fitness and nutrition assistance.

17. Team Contribution

Santhosh C – Team Leader and project implementation/documentation. Tharun M.P – Team Member. Rajiv P – Team Member. Logesh I – Team Member. The project was developed collaboratively as a Naan Mudhalvan team project.

18. Conclusion

FitSense AI demonstrates how Artificial Intelligence can be combined with a secure REST API and database for personalized fitness management. The system provides authentication, workout CRUD operations, search, AI workout recommendations, and AI fitness insights, while providing a foundation for future frontend development.

19. References

- Node.js Documentation – <https://nodejs.org/>

- Express.js Documentation – <https://expressjs.com/>
- MongoDB Documentation – <https://www.mongodb.com/docs/>
- Google Gemini API Documentation – <https://ai.google.dev/>
- Postman Documentation – <https://learning.postman.com/>